

BHABHA UNIVERSITY, BHOPAL

NH-12, Hoshangabad Road, Jatkhedhi, Bhopal-462026

DIPLOMA IN COMPUTER APPLICATION (DCA)



**DETAILED SYLLABUS
(EFFECTIVE FROM JULY 2018)**

**Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL**

BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN COMPUTER APPLICATION (DCA)

(EFFECTIVE FROM JULY 2018)

DETAILED SYLLABUS

SCHEME

SEMESTER -I

Subject Code	Subject Name	Scheme			Theory Paper	Internal Evaluation	Practical Exams	Total Marks
		L	T	P				
D101	Fundamentals of Computers	4			70	30		100
D102	PC Packages (Windows, MS Word, MS Excel, MS Powerpoint)	4	1	3	70	30	25	125
D103	Elective - 1 A-MS Access B-DBMS	4		3	70	30	25	125
Semester Total 350								

(*L-Lecture, T-Tutorial, P-Practical)

SCHEME

Subject Code	Subject Name	Scheme			Theory Paper	Internal Evaluation	Practical Exams	Total Marks
		L	T	P				
D201	IT Trends and Technologies	4			70	30		100
D202	Internet and Web Page Designing	4	1	3	70	30	25	125
D203(A) D203(B)	Elective – 2 Corel Draw Desk Top Publishing (Page Maker, Photoshop)	4		3	70	30	25	125
Semester Total 350								

BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN COMPUTER APPLICATION (DCA)

DETAILED SYLLABUS

Fundamentals of Computers - D101

- UNIT I:** Brief history of computers, Computer System Concept, Computer System development, characteristics, Capabilities and Limitations, Types of Computers, Personal Computer (PCs) - IBM PCs, Types of PCs- Desktop, Laptop, Notebook, Palmtop, etc. Basic Components of a Computer System - Control Unit, ALU, Input/output, semiconductor Memory. Storage fundamentals - Primary Vs Secondary memory.
- UNIT II :** Input devices: keyboard, mouse, Joystick, Scanners, Digital Camera, MICR, OCR, OMR, , Light pen, touch Monitors - Characteristics and types of monitor, Size, Resolution, Refresh Rate, Dot Pitch.
Output devices: Video Standard - VGA, SVGA, XGA etc. Printers - Daisy wheel, Dot Matrix, Inkjet, Laser. Plotter, Sound Card and Speakers.
- UNIT III :** Various Storage Devices - Magnetic Disks, Hard Disk Drives, Floppy, Disks, Optical Disks, Computer Software Need, Types of Software's - System software, Application software System Software - Operating System, compiler Assemblers, Interpreter.
- UNIT IV :** Operating Systems –Functions ,Types- Batch, Single, Multiprogramming, Multiprocessing Programming languages- Machine, Assembly, High Level, 4GL, Application Software - Word-processing, Spreadsheet, Presentation Graphics, Data Base Management Software, **Number System**, Number System of computers- Binary, Octal, Hexadecimal, their conversion. Coding System – ASCII, EBCDIC.
- UNIT V :** Direction of Transmissions Flow-Simplex, Half Duplex Full Duplex, Types of Network - LAN, WAN, MAN etc. Topologies of LAN - Ring, Bus, Star, Mesh and Tree topologies.
Computer Virus: Virus working principals, Types of viruses, Virus detection and Prevention Viruses on network.

TEXT & REFERENCE BOOKS :

COMPUTERS TODAY BY S.K. BASANDRA, GALGOTIA PUBLICATIONS

FUNDAMENTALS OF INFORMATION TECHNOLOGY BY ALEXIS LEON & MATHEWS LEON, VIKAS
PUBLISHING HOUSE, NEW DELHI.

DOS QUICK REFERENCE BY RAJEEV MATHUR, GALGOTIA PUBLICATIONS

BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN COMPUTER APPLICATION (DCA)

DETAILED SYLLABUS

PC Packages (Windows, MS Word, MS Excel, MS Powerpoint)- D102

- UNIT I:** **MS Windows:** Introduction to MS Windows; Features of Windows; Various versions of Windows & its use; Working with Windows; My Computer & Recycle bin ; Desktop, Icons and Windows Explorer; Screen description & working styles of Windows; Dialog Boxes & Toolbars; Working with Files & Folders; simple operations like copy, delete, moving of files and folders from one drive to another, Shortcuts & Autostarts; Accessories and Windows Settings using Control Panel- setting common devices using control panel, modem, printers, audio, network, fonts, creating users, internet settings, Start button & Program lists; Installing and Uninstalling new Hardware & Software program on your computer.
- UNIT II:** Office activates and their software requirements, Word-processing, Spreadsheet, Presentation graphics, Database, introduction and comparison of various office suites like MS-Office, Lotus-Office, Star-Office, Open-Office etc. **MS Word Basics:** Introduction to MS Office, Introduction to MS Word, Features & area of use. Working with MS Word, Menus & Commands, Toolbars & Buttons, Shortcut Menus, Wizards & Templates, Creating a New Document, Different Page Views and layouts, Applying various Text Enhancements, Working with -Styles, Text Attributes, Paragraph and Page Formatting, Text Editing using various features ; Bullets, Numbering, Auto formatting, Printing & various print options.
- UNIT III:** **Advanced Features of MS-Word:** Spell Check, Thesaurus, Find & Replace; Headers & Footers, Inserting - Page Numbers, Pictures, Files, Auto texts, Symbols etc., Working with Columns, Tabs & Indents, Creation & Working with Tables including conversion to and from text, Margins & Space management in Document, Adding References and Graphics, Mail Merge, Envelops & Mailing Labels. Importing and exporting to and from various formats.
- UNIT IV:** **MS Excel:** Introduction and area of use, Working with MS Excel, concepts of Workbook & Worksheets, Using Wizards, Various Data Types, Using different features with Data, Cell and Texts, Inserting, Removing & Resizing of Columns & Rows, Working with Data & Ranges, Different Views of Worksheets, Column Freezing, Labels, Hiding, Splitting etc., Using different

features with Data and Text; Use of Formulas, Calculations & Functions, Cell Formatting including Borders & Shading, Working with Different Chart Types; Printing of Workbook & Worksheets with various options.

UNIT V: **MS PowerPoint:** Introduction & area of use, Working with MS PowerPoint, Creating a New Presentation, Working with Presentation, Using Wizards; Slides & its different views, Inserting, Deleting and Copying of Slides; Working with Notes, Handouts, Columns & Lists, Adding Graphics, Sounds and Movies to a Slide; Working with PowerPoint Objects, Designing & Presentation of a Slide Show, Printing Presentations, Notes, Handouts with print options.

Outlook Express: Features and uses, Configuration and using Outlook Express for accessing e-mails in office.

TEXT & REFERENCE BOOKS:

WINDOWS XP COMPLETE REFERENCE. BPB PUBLICATIONS

MS OFFICE XP COMPLETE BPB PUBLICATION

MS WINDOWS XP HOME EDITION COMPLETE, BPB PUBLICATION.

JOE HABRAKEN, MICROSOFT OFFICE 2000, 8 IN 1, BY, PRENTICE HALL OF INDIA

I.T .TOOLS AND APPLICATIONS, BY A. MANSOOR, PRAGYA PUBLICATIONS, MATURA

BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN COMPUTER APPLICATION (DCA)

DETAILED SYLLABUS

Elective – 1, D103(A) MS Access

- UNIT I:** Introduction to database -What is a Database, Why use a Relational Database, concept of primary key relationship Introduction to MS Access (Objects, Navigation).
- UNIT-II:** Create a Table in MS Access -Data Types, Field Properties, Fields: names, types, properties-- default values, format, caption, validation rules Data Entry, Add record, delete record and edit text, Sort, find/replace, Filter / select, rearrange columns, freeze columns. Edit a Tables- copy, delete, import, modify table structure, find, replace.
- UNIT-III:** Add a relationship, set a rule for Referential Integrity, change the join type, delete a relationship, save relationship Queries & Filter - difference between queries and filter, filter using multiple fields ,Create Query with one table, Find record with select query, find duplicate record with query, find unmatched record with query, run query, save and change query.
- UNIT-IV:** Introduction to Forms Types of Basic Forms: Columnar, Tabular, Datasheet, Main/ Sub forms, add headers and footers, add fields to form, add text to form use label option button, check box, combo box, list box Forms Wizard, Create Template.
- UNIT-V:** Introduction to Reports, Types of Basic Reports: Single Column, Tabular Report Groups/Total, single table report, multi table report preview report print report, Creating Reports and Labels, Wizard.

TEXT & REFERENCE BOOKS:

MS OFFICE XP COMPLETE BPB PUBLICATION ISBN 8 1-7656-564-4

MS ACCESS FAST & EASY BY FAITHE WEMPEN PHI .

BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN COMPUTER APPLICATION (DCA)
DETAILED SYLLABUS

Elective – 1, D103(B) DBMS

- UNIT–I:** Introduction to database -What is a Database, Why use a Relational Database, concept of primary key relationship, Introduction to MS Access (Objects, Navigation).
- UNIT–II:** Create a Table in MS Access -Data Types, Field Properties, Fields: names, types, properties-- default values, format, caption, validation rules Data Entry, Add record, delete record and edit text, Sort, find/replace, filter/ select, rearrange, columns, freeze columns. Edit a Tables- copy, delete, import, modify table structure, find, replace.
- UNIT–III:** Add a relationship, set a rule for Referential Integrity, change the join type, delete a relationship, save relationship Queries & Filter - difference between queries and filter, filter using multiple fields ,Create Query with one table, find record with select query, find duplicate record with query, find unmatched record with query, run query, save and change query.
- UNIT–IV:** Introduction to Forms Types of Basic Forms: Columnar, Tabular, Datasheet, Main/Sub forms, add headers and footers, add fields to form, add text to form use label option button, check box, combo box, list box Forms Wizard, create templates.
- UNIT–V:** Introduction to Reports, Types of Basic Reports: Single Column, Tabular Report Groups/Total, single table report, multi table report preview report print report, Creating Reports and Labels, Wizard.

TEXT & REFERENCE BOOKS:

MS OFFICE XP COMPLETE BPB PUBLICATION ISBN 8 1-7656-564-4

MS ACCESS FAST & EASY BY FAITHE WEMPEN PHI .

BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN COMPUTER APPLICATION (DCA)
DETAILED SYLLABUS

IT Trends and Technologies D201

- UNIT-I:** Introduction to Distributed system, Advantage and Disadvantage of Distributed System, Artificial Intelligence and Expert system- Concepts of AI & Expert Systems, Merits and Demerits of Expert system, Application of Expert system and AI.
- UNIT-II:** Introduction to Virtual Reality and multimedia: Introduction, Brief History of virtual reality, Present uses of virtual reality, various software of multimedia.
- UNIT-III:** E-Commerce: an Introductions, Concepts, Advantages and Disadvantages, Technology in E-Commerce, Benefits and impact of e-commerce, Electronic Payment Systems: Introduction, Types of Electronic Payment Systems.
- UNIT-IV:** CELLULAR TRANSMISSION: Radio Frequency Communications, Wireless Services Wireless & Mobile Computing, Cellular System Cell, Mobile Switching office, Hands off, Base Station, Frequency Reuse and Cluster.
- UNIT-V:** E-governance, e-democracy, Government efforts to encourage citizen participation Privacy and Security Issues, Information security management Digital Divide, Introduction of Cyber Crime, various website for e- governance.

TEXT & REFERENCE BOOKS:

FUNDAMENTALS OF INFORMATION TECHNOLOGY BY ALEX LEON & MLEON

FRONTIERS OF ELECTRONIC COMMERCE, BY- KALAKOTA, RAVI; STONE, TOM; WHINSTON, ANDREW B, ADDISON WESLEY PUBLISHING CO, ISBN 8178080575.

E-COMMERCE AN INDIAN PERSPECTIVE (SECOND EDITION) BY PT JOSEPH, S.J. PRENTICE-HALL OF INDIA

RECENT MAGAZINES OF COMPUTERS AND COMMUNICATION

BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN COMPUTER APPLICATION (DCA)
DETAILED SYLLABUS

Internet and Web Page Designing D202

- UNIT-I:** Applications of Internet, History of Internet, WWW, Various Services , World Wide Web (WWW)
History, Working, Web Browsers, Its function Concept of Search Engines, client server architecture .
- UNIT-II:** **Internet :** Evolution, Protocols, Interface Concepts, Internet Vs Internet, Growth of Internet, ISP,
Connectivity - Dial-up, Leased line, VSAT etc., URLs, Domain names, Portals, Applications.
E-Mail : Concepts , Basics of Sending & Receiving, E-mail, Free E-mail services.
- UNIT-III:** Transfer Protocols, Telnet & Chatting , Client/Server Architecture Characteristic, FTP & its usages.
Telnet Concept, Remote Logging, Protocols, Internet chatting - Voice chat, text chat.
- UNIT-IV:** Searching the Web, HTTP, URLs, Web Servers, Web Protocols. Web Publishing Concepts, Domain
Name Registration, . HTML, Design Tools, HTML Editors , Image Editors.
- UNIT-V:** HTML Concepts of Hypertext, Versions of HTML, Elements of HTML Syntax, Head & Body Sections,
Building HTML Documents, Inserting Texts, Images, Hyperlinks, Backgrounds And Colour Controls,
Different HTML Tags, Table Layout and Presentation, Use of Font Size & Attributes, List types and its
Tags.

TEXT & REFERENCE BOOKS:

LEVEL MODULE - M 1.2 - INTERNET & WEB PAGE DESIGNING BY V.K.JAIN BPB PUBLICATIONS

INTERNET FOR EVERYONE - ALEXIS LEON AND MATHEWS LEON, VIKAS PUBLICATIONS

INTERNET FOR DUMMIES - PUSTAK MAHAL, NEW DELHI

A BEGINNER'S GUIDE TO HTML

[HTTP://WWW.NCSA.NINE.EDIT/GENERAL/INTERNET/WWW/HTML.PRMTER](http://WWW.NCSA.NINE.EDIT/GENERAL/INTERNET/WWW/HTML.PRMTER)

BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN COMPUTER APPLICATION (DCA)
DETAILED SYLLABUS

Elective – 2 D203 (A) Corel Draw

- UNIT–I:** Introduction to CorelDraw, Use and importance in Designing, Various Graphic Files and File Extensions, Vector Image and Raster Images , Introduction to Screen and Work Area.
- UNIT–II:** Introduction to Tools of CorelDraw, Managing Palettes ,Working with Images, Patterns and Textures, Working with Shapes, Colours and Fills ,Image Rasterisation and Editing, Transformation Menu.
- UNIT–III:** Coreldraw Files and supporting documents, Import and Export of Files and File formats Page Setup and Designing, Using Styles and Templates, Working with Text, Formatting Text, Text Attributes.
- UNIT–IV:** Designing Different Page Layouts, Column Layout, Working with Layers., Special Effect to Objects and Texts, Contour Tool , Layout for News Paper and Magazines.
- UNIT–V:** Preparation of Visiting Cards & Invitation Cards, Shaping Dockers & Logo Design Introduction Brochure & Books, Introduction to Magazine Designing.

TEXT & REFERENCE BOOKS:

CORELDRAW X4 FOR SIMPLE STEPS

CORELDRAW X4 THE OFFICIAL GUIDE BY GARY DAVID BOUTON

BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN COMPUTER APPLICATION (DCA)
DETAILED SYLLABUS

Elective – 2 DCA 203(B) Desk Top Publishing (Page Maker, Photoshop)

UNIT-I: D.T.P For Publications Introduction to Printing, Types of Printing, Offset Printing, Working of offset Printing, Transparent Printout, Negative & Positives for Plate were making, Use of Desk Top Publishing in Publications, Importance of D.T.P in Publication, Advantage of D.T.P in Publication, Mixing of graphics & Image in a single page production, Laser printers - Use, Types, Advantage of lager printer in publication.

UNIT-II: Page Layout Different page format / Layouts, News paper page format, Page orientations, Columns & Gutters, Printing in reduced sizes.

Introductions To Page Maker Page Maker Icon and help, Tool Box, Styles, Menus etc., Different screen Views, Importing text/Pictures, Auto Flow, Columns.

UNIT-III: Master Pages and Stories, Story Editor, Menu Commands and short-cut commands, Spell check, Find & Replace, Import Export etc., Fonts, Points Sizes, Spacing etc., Installing Printers, Scaling (Percentages), Printer setup.

UNIT-IV: Introduction to Adobe Photoshop & Documents, Various Graphic Files and Extensions, Vector Image and Raster Images, Various Colour Modes and Models.

UNIT-V: Introduction to Screen and Work Area, Photoshop Tools & Palettes, Use of Layers & Filters, Working with Images.

TEXT & REFERENCE BOOKS:

PAGE MAKER 4.0 & 5.0 BY B.P.O. PUBLICATIONS.

PRAKHAR COMPLETE COURSE FOR DTP (CORELDRAW, PAGEMAKER, PHOTOSHOP)